



7893 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
CT286				
	1		NIRCam Imaging	(1) CT286
	2		NIRSpec Fixed Slit Spectroscopy	(1) CT286
DES0228				
	3		NIRCam Imaging	(2) DES0228
	4		NIRSpec Fixed Slit Spectroscopy	(2) DES0228
DES0033				
	5		NIRCam Imaging	(3) DES0033
	6		NIRSpec Fixed Slit Spectroscopy	(3) DES0033
S5-0836+71				
	7		NIRCam Imaging	(4) S5-0836+71
	8		NIRSpec Fixed Slit Spectroscopy	(4) S5-0836+71
SBS1425+606				
	9		NIRCam Imaging	(5) SBS1425+606
	10		NIRSpec Fixed Slit Spectroscopy	(5) SBS1425+606
RMID363				
	11		NIRCam Imaging	(6) RM363

Folder	Observation	Label	Observing Template	Science Target
	12		NIRSpec Fixed Slit Spectroscopy	(6) RM363
RMID372				
	13		NIRCam Imaging	(7) RM372
	14		NIRSpec Fixed Slit Spectroscopy	(7) RM372

ABSTRACT

A key constraint on the co-evolution scenario of supermassive black holes and their host galaxies is the observed correlations between the black hole (BH) mass and host stellar properties at Cosmic Noon ($z \sim 2$), where both quasar activity and global star formation reached their peak. Such constraints rely on reliable measurements of both BH masses and host stellar properties, and there is currently no consensus on whether or not such correlations exist at $z \sim 2$. This program will observe a sample of seven of the most massive quasars at $z \sim 2-3$ that have direct reverberation mapping-based BH masses, using NIRCam $\sim 1-5$ micron imaging and NIRSpec Fixed Slit spectroscopy. With these data, we will measure host stellar properties in these luminous quasars at Cosmic Noon. The proposed sample will significantly expand the dynamic range in BH mass (by a factor of ~ 4) beyond existing samples to facilitate the measurement of BH-host scaling relations. This program will provide a benchmark sample for comparative studies of massive galaxies and luminous quasars across cosmic time.

OBSERVING DESCRIPTION

We will perform NIRCam imaging and NIRSpec FS spectroscopy for seven $z \sim 2-3$ quasars with direct black hole mass estimates based on reverberation mapping. The NIRCam imaging is primarily used to measure morphology, local environment, and stellar populations of the hosts upon imaging decomposition in multiple bands. The NIRSpec FS spectroscopy is primarily used to measure the stellar velocity dispersion.

Each target will have two observations, and each observation can be fit in a single visit. All targets can be considered compact sources (with sizes $< 1''$) in sparse extragalactic fields. All targets are bright, with J-band magnitude of 15-19 (Vega).

The first visit/observation is NIRCam imaging in two pairs of short+long-wave filters, which are optimized according to source redshift: F090W+F277W and F150W+F444W for three objects with $z < \sim 2$, F115W+F277W and F200W+F444W for three objects with $z \sim 2.5$, and F150W+F356W and F200W+F444W for the remaining object with $z = 3.2$. NIRCam FULL subarrays with both Modules A and B are adopted for all objects to increase the FoV for better PSF modeling using nearby field stars. We adopt RAPID/BRIGHT1 readout pattern for five targets to reduce saturation in central pixels. ETC is run in each filter separately using the template target spectrum, individual redshift and observed JHKW1W2 photometry for each target to avoid pixel saturation and SNR estimation. For each target, we will adopt a 4-point STANDARD subpixel dither

JWST Proposal 7893 (Created: Friday, May 23, 2025, 6:00:10PM Eastern Standard Time) - Overview

pattern and a total exposure time of ~ 10 minutes (both filter pairs combined) for five bright sources and ~ 35 minutes for two faint sources to reach a SNR of $>\sim 30$ for host emission within $0.3''$ radius. This moderate science exposure time will provide sufficient depth for reliable host detection. This dither pattern is sufficient to reach Nyquist sampling of the PSF for all filters but F090W, whose PSF is still slightly undersampled. This compromise is acceptable since the F090W filter samples the rest-frame UV and is not primarily used for host morphological measurements and stellar mass estimation. Increasing the number of dither points will lead to extra overhead with minimal improvements for our primary science goals. We will use field stars in all NIRCcam pointings for the targets to construct optimal PSF models for our quasar/host decomposition.

The second visit/observation is the NIRSpec FS spectroscopy. We will use the G140M/F100LP for one target with $z=1.7$ and G235M/F170LP for the remaining six targets. We will adopt WATA with the science target itself for target acquisition. We will use S200A1 slit with a two-point primary dither (nodding) plus two-point subpixel spatial dither pattern. To meet the SNR requirement for reliable stellar velocity dispersion measurement, we would need ~ 20 min total exposure for five bright sources and ~ 60 min for the remaining two faint sources. Saturation is not an issue for the proposed NIRSpec observations.

Proposal 7893 - Targets - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	CT286	RA: 10 17 23.9162 (154.3496508d) Dec: -20 46 58.50 (-20.78292d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				
(2)	DES0228	RA: 02 28 28.2018 (37.1175075d) Dec: -04 00 44.42 (-4.01234d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				
(3)	DES0033	RA: 00 33 52.7363 (8.4697346d) Dec: -42 54 52.66 (-42.91463d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				
(4)	S5-0836+71	RA: 08 41 24.3653 (130.3515221d) Dec: +70 53 42.17 (70.89505d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				
(5)	SBS1425+606	RA: 14 26 56.1780 (216.7340750d) Dec: +60 25 50.87 (60.43080d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				
(6)	RM363	RA: 14 21 13.3000 (215.3054167d) Dec: +52 49 29.90 (52.82497d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				
(7)	RM372	RA: 14 12 36.4800 (213.1520000d) Dec: +54 01 52.10 (54.03114d) Equinox: J2000	Proper Motion RA: 0 Proper Motion Dec: 0	
Comments: Category=Galaxy Description=[Quasars] Extended=NO				

Proposal 7893 - Observation 1 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 1										Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous	
	(1)	CT286	RA: 10 17 23.9162 (154.3496508d) Dec: -20 46 58.50 (-20.78292d) Equinox: J2000				Proper Motion RA: 0 Proper Motion Dec: 0				
	Comments: Category=Galaxy Description=[Quasars] Extended=NO										
Template	Module		Subarray					Target Placement			
	ALL		FULL					Module B (B4 corner)			
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F277W	BRIGHT1	5	1	4	4	386.524		
	2	F200W	F444W	BRIGHT1	5	1	4	4	386.524		
Special Requirements	Aperture PA Range 0.05262691 to 16.05262691 Degrees (V3 0.0 to 16.0) Aperture PA Range 34.05262691 to 36.05262691 Degrees (V3 34.0 to 36.0) Aperture PA Range 40.05262691 to 62.05262691 Degrees (V3 40.0 to 62.0) Aperture PA Range 93.05262691 to 154.05262691 Degrees (V3 93.0 to 154.0) Aperture PA Range 164.05262691 to 169.05262691 Degrees (V3 164.0 to 169.0) Aperture PA Range 181.05262691 to 253.05262691 Degrees (V3 181.0 to 253.0) Aperture PA Range 261.05262691 to 279.05262691 Degrees (V3 261.0 to 279.0) Aperture PA Range 289.05262691 to 359.05262691 Degrees (V3 289.0 to 359.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 1, 2, Non-interruptible										

Proposal 7893 - Observation 2 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 2											Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	CT286	RA: 10 17 23.9162 (154.3496508d)				Proper Motion RA: 0					
			Dec: -20 46 58.50 (-20.78292d)				Proper Motion Dec: 0					
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	227049.80	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	2							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRSIRS2RAPID	20	1	1	NONE	4	4	1225.467	

Proposal 7893 - Observation 2 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 7893 - Observation 3 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 3										Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(2)	DES0228	RA: 02 28 28.2018 (37.1175075d) Dec: -04 00 44.42 (-4.01234d) Equinox: J2000			Proper Motion RA: 0 Proper Motion Dec: 0					
	Comments: Category=Galaxy Description=[Quasars]										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B (B4 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F090W	F277W	BRIGHT1	5	1	4	4	386.524		
	2	F162M+F150W2	F444W	BRIGHT1	5	1	4	4	386.524		
Special Requirements	Aperture PA Range 0.05262691 to 92.05262691 Degrees (V3 0.0 to 92.0) Aperture PA Range 98.05262691 to 102.05262691 Degrees (V3 98.0 to 102.0) Aperture PA Range 104.05262691 to 140.05262691 Degrees (V3 104.0 to 140.0) Aperture PA Range 146.05262691 to 150.05262691 Degrees (V3 146.0 to 150.0) Aperture PA Range 152.05262691 to 181.05262691 Degrees (V3 152.0 to 181.0) Aperture PA Range 187.05262691 to 359.05262691 Degrees (V3 187.0 to 359.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 3, 4, Non-interruptible										

Proposal 7893 - Observation 4 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 4										Fri May 23 23:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	DES0228	RA: 02 28 28.2018 (37.1175075d)			Proper Motion RA: 0						
			Dec: -04 00 44.42 (-4.01234d)			Proper Motion Dec: 0						
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	227049.84	
Template	HFF Readout Mode			Slit				Subarray				
	false			S200A1				FULL				
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	2						SPATIAL				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRSIRS2RAPID	20	1	1	NONE	4	4	1225.467	

Proposal 7893 - Observation 4 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 7893 - Observation 5 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 5									Fri May 23 23:00:10 GMT 2025	
	Diagnostic Status: Warning										
Observing Template: NIRCam Imaging											
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(3)	DES0033	RA: 00 33 52.7363 (8.4697346d)			Proper Motion RA: 0					
			Dec: -42 54 52.66 (-42.91463d)			Proper Motion Dec: 0					
			Equinox: J2000								
Template	Comments: Category=Galaxy Description=[Quasars] Extended=NO										
	Module	Subarray				Target Placement					
	ALL	FULL				Module B (B4 corner)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F277W	BRIGHT1	8	1	4	4	644.206		
	2	F200W	F444W	BRIGHT2	4	1	4	4	343.577		
Special Requirements	Aperture PA Range 4.05262691 to 82.05262691 Degrees (V3 4.0 to 82.0) Aperture PA Range 109.05262691 to 120.05262691 Degrees (V3 109.0 to 120.0) Aperture PA Range 128.05262691 to 237.05262691 Degrees (V3 128.0 to 237.0) Aperture PA Range 245.05262691 to 339.05262691 Degrees (V3 245.0 to 339.0) Aperture PA Range 357.05262691 to 358.05262691 Degrees (V3 357.0 to 358.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 5, 6, Non-interruptible										

Proposal 7893 - Observation 6 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 6											Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	DES0033	RA: 00 33 52.7363 (8.4697346d)				Proper Motion RA: 0					
			Dec: -42 54 52.66 (-42.91463d)				Proper Motion Dec: 0					
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB2048	F140X	NRSRAPID	3	1	1	3.628	227049.86	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	2							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRSIRS2RAPID	20	1	1	NONE	4	4	1225.467	

Proposal 7893 - Observation 6 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 5, 6, Non-interruptible
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Proposal 7893 - Observation 7 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 7										Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(4)	S5-0836+71	RA: 08 41 24.3653 (130.3515221d)			Proper Motion RA: 0					
			Dec: +70 53 42.17 (70.89505d)			Proper Motion Dec: 0					
	Equinox: J2000										
	Comments: Category=Galaxy Description=[Quasars] Extended=NO										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B (B4 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F277W	BRIGHT1	5	1	4	4	386.524		
	2	F162M+F150W2	F430M	BRIGHT1	5	1	4	4	386.524		
Special Requirements	Aperture PA Range 0.05262691 to 41.05262691 Degrees (V3 0.0 to 41.0) Aperture PA Range 49.05262691 to 59.05262691 Degrees (V3 49.0 to 59.0) Aperture PA Range 67.05262691 to 72.05262691 Degrees (V3 67.0 to 72.0) Aperture PA Range 80.05262691 to 91.05262691 Degrees (V3 80.0 to 91.0) Aperture PA Range 98.05262691 to 174.05262691 Degrees (V3 98.0 to 174.0) Aperture PA Range 184.05262691 to 311.05262691 Degrees (V3 184.0 to 311.0) Aperture PA Range 316.05262691 to 359.05262691 Degrees (V3 316.0 to 359.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 7, 8, Non-interruptible										

Proposal 7893 - Observation 8 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 8											Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	S5-0836+71	RA: 08 41 24.3653 (130.3515221d)				Proper Motion RA: 0					
			Dec: +70 53 42.17 (70.89505d)				Proper Motion Dec: 0					
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPIDD6	3	1	1	0.26	227049.88	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	2							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRSIRS2RAPID	20	1	1	NONE	4	4	1225.467	

Proposal 7893 - Observation 8 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 7, 8, Non-interruptible
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Proposal 7893 - Observation 9 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 9										Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(5)	SBS1425+606	RA: 14 26 56.1780 (216.7340750d) Dec: +60 25 50.87 (60.43080d) Equinox: J2000			Proper Motion RA: 0 Proper Motion Dec: 0					
	Comments: Category=Galaxy Description=[Quasars] Extended=NO										
Template	Module	Subarray				Target Placement					
	ALL	FULL				Module B (B4 corner)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F150W	F356W	BRIGHT1	4	1	4	4	300.63		
	2	F182M	F460M	BRIGHT1	4	1	4	4	300.63		
Special Requirements	Aperture PA Range 0.05262691 to 64.05262691 Degrees (V3 0.0 to 64.0) Aperture PA Range 72.05262691 to 75.05262691 Degrees (V3 72.0 to 75.0) Aperture PA Range 80.05262691 to 214.05262691 Degrees (V3 80.0 to 214.0) Aperture PA Range 221.05262691 to 230.05262691 Degrees (V3 221.0 to 230.0) Aperture PA Range 241.05262691 to 242.05262691 Degrees (V3 241.0 to 242.0) Aperture PA Range 249.05262691 to 261.05262691 Degrees (V3 249.0 to 261.0) Aperture PA Range 267.05262691 to 359.05262691 Degrees (V3 267.0 to 359.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 9, 10, Non-interruptible										

Proposal 7893 - Observation 10 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 10										Fri May 23 23:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(5)	SBS1425+606	RA: 14 26 56.1780 (216.7340750d)				Proper Motion RA: 0					
			Dec: +60 25 50.87 (60.43080d)				Proper Motion Dec: 0					
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	227049.90	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	2							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRSIRS2RAPID	20	1	1	NONE	4	4	1225.467	

Proposal 7893 - Observation 10 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 9, 10, Non-interruptible
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Proposal 7893 - Observation 11 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 11										Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(6)	RM363	RA: 14 21 13.3000 (215.3054167d)			Proper Motion RA: 0					
			Dec: +52 49 29.90 (52.82497d)			Proper Motion Dec: 0					
			Equinox: J2000								
	Comments: Category=Galaxy Description=[Quasars] Extended=NO										
Template	Module	Subarray				Target Placement					
	ALL	FULL				Module B (B4 corner)					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F115W	F277W	SHALLOW4	5	1	4	4	1030.73		
	2	F200W	F444W	SHALLOW4	5	1	4	4	1030.73		
Special Requirements	Aperture PA Range 0.05262691 to 53.05262691 Degrees (V3 0.0 to 53.0) Aperture PA Range 62.05262691 to 71.05262691 Degrees (V3 62.0 to 71.0) Aperture PA Range 81.05262691 to 82.05262691 Degrees (V3 81.0 to 82.0) Aperture PA Range 86.05262691 to 87.05262691 Degrees (V3 86.0 to 87.0) Aperture PA Range 90.05262691 to 302.05262691 Degrees (V3 90.0 to 302.0) Aperture PA Range 308.05262691 to 359.05262691 Degrees (V3 308.0 to 359.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 11, 12, Non-interruptible										

Proposal 7893 - Observation 12 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 12											Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning											
	Observing Template: NIRSspec Fixed Slit Spectroscopy											
Diagnostics	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(6)	RM363	RA: 14 21 13.3000 (215.3054167d)				Proper Motion RA: 0					
			Dec: +52 49 29.90 (52.82497d)				Proper Motion Dec: 0					
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB2048	CLEAR	NRSRAPID	3	1	1	3.628	227049.92	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	2							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G235M/F170LP	S200A1	NRSIRS2RAPID	61	1	1	NONE	4	4	3618.045	

Proposal 7893 - Observation 12 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 11, 12, Non-interruptible
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Proposal 7893 - Observation 13 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 13										Fri May 23 23:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	RM372	RA: 14 12 36.4800 (213.1520000d)			Proper Motion RA: 0					
			Dec: +54 01 52.10 (54.03114d)			Proper Motion Dec: 0					
	Equinox: J2000										
	Comments: Category=Galaxy Description=[Quasars] Extended=NO										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Module B (B4 corner)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				4	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F090W	F277W	SHALLOW4	6	1	4	4	1245.465		
	2	F150W	F444W	SHALLOW4	5	1	4	4	1030.73		
Special Requirements	Aperture PA Range 0.05262691 to 170.05262691 Degrees (V3 0.0 to 170.0) Aperture PA Range 183.05262691 to 184.05262691 Degrees (V3 183.0 to 184.0) Aperture PA Range 188.05262691 to 188.05262691 Degrees (V3 188.0 to 188.0) Aperture PA Range 196.05262691 to 220.05262691 Degrees (V3 196.0 to 220.0) Aperture PA Range 226.05262691 to 238.05262691 Degrees (V3 226.0 to 238.0) Aperture PA Range 244.05262691 to 359.05262691 Degrees (V3 244.0 to 359.0) Offset 38.0 arcsec, 38.0 arcsec Fiducial Point Override NRCBS_FULL										
	Sequence Observations 13, 14, Non-interruptible										

Proposal 7893 - Observation 14 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Observation	Proposal 7893, Observation 14										Fri May 23 23:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(7)	RM372	RA: 14 12 36.4800 (213.1520000d)				Proper Motion RA: 0					
			Dec: +54 01 52.10 (54.03114d)				Proper Motion Dec: 0					
			Equinox: J2000									
	Comments: Category=Galaxy Description=[Quasars] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB2048	CLEAR	NRSRAPIDD6	3	1	1	14.452	227049.94	
Template	HFF Readout Mode				Slit				Subarray			
	false				S200A1				FULL			
Dithers	#	Primary Dither Positions							Sub-Pixel Pattern			
	1	2							SPATIAL			
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G140M/F100LP	S200A1	NRSIRS2RAPID	61	1	1	NONE	4	4	3618.045	

Proposal 7893 - Observation 14 - Host Galaxy Properties of the Most Massive Quasars at Cosmic Noon

Special Requirements	Sequence Observations 13, 14, Non-interruptible
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